Bihar Engineering University, Patna

B.Tech 4th Semester Examination, 2025

Course: B.Tech
Code: 101402

Subject: Disaster Preparedness & Planning

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) NDRF stands for
- (i) National Disaster Rescue Force
- (ii) National Disaster Response Force
- (iii) National Defence Rescue Force
- (iv) National Development Response Force

(b) A human stampede is an example of a
- (i) Natural disaster
- (ii) Socio-natural disaster
- (iii) Man-made disaster
- (iv) Biological disaster

(c) The Disaster Management Cycle typically consists of how many major phases?
- (i) 2
- (ii) 3
- (iii) 4
- (iv) 6

(d) Which of the following is a *slow-onset* natural disaster?
- (i) Earthquake
- (ii) Tsunami
- (iii) Drought
- (iv) Flash flood

(e) Damage and loss assessment (DALA) is performed in the
- (i) Response phase
- (ii) Preparedness phase
- (iii) Mitigation phase
- (iv) Recovery phase

(f) The State Disaster Management Authority (SDMA) is headed by
- (i) Governor
- (ii) Chief Minister
- (iii) Home Minister of the state
- (iv) Chief Secretary

(g) Public awareness during disasters is the responsibility of
- (i) Only NGOs
- (ii) Only media
- (iii) All levels of government
- (iv) Private companies

(h) Sustainable reconstruction after a disaster emphasizes
- (i) Quick rebuilding without planning
- (ii) Rebuilding using environmentally friendly materials and designs
- (iii) Rebuilding on the same hazard-prone site
- (iv) Ignoring environmental impact

(i) The concept of "Build Back Better" emphasizes
- (i) Faster construction without environmental assessment
- (ii) Rebuilding stronger, safer, and more sustainably
- (iii) Rebuilding identical to pre-disaster condition
- (iv) Lower quality housing for affordability

(j) Unplanned urbanization primarily increases vulnerability by
- (i) Increasing open spaces
- (ii) Creating high population density and poor infrastructure
- (iii) Enhancing natural drainage
- (iv) Improving building code compliance

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Q.2	(a)	What is a disaster management cycle? Explain this cycle in detail.	[7]
	(b)	When can developmental activities become the cause of disasters?	[7]
Q.3	(a)	Describe any flooding disaster that occurred in India and write the various lessons that you have learned from it?	[7]
	(b)	Write in detail about earthquake vulnerability assessment and building and infrastructure?	[7]
Q.4	(a)	Classify different types of disasters. How disaster can influence the development?	[7]
	(b)	Explain the community's role in disaster preparedness, response, and recovery. Why is community participation crucial in disaster management?	[7]
Q.5	(a)	Discuss the contribution of NDMA in capacity building, training, and awareness generation.	[7]
	(b)	Describe major national DRR programmes implemented in India for reducing vulnerability to natural disasters.	[7]
Q.6	(a)	Define mitigation? Discuss structural and non-structural mitigation measures	[7]
	(b)	Discuss the importance of disaster preparedness plans at national, state, district, and community levels	[7]
Q.7	(a)	Explain the concept of hazard, exposure, vulnerability, and capacity.	[7]
	(b)	Discuss the role of technology to remote sensing, GIS, satellites in early warning systems.	[7]
Q.8	(a)	"Disaster risk is a function of hazard, vulnerability, and capacity." Explain with examples.	[7]
	(b)	Describe the importance of academic and research institutions in disaster risk assessment and capacity building.	[7]
Q.9	(a)	Differentiate between preparedness and mitigation.	[7]
	(b)	Explain the objectives and key guidelines of the Disaster Management Act, 2005.	[7]

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